

Aml HMAS: A Hypermedia MAS for Goal-Driven Interactions with every-day Smart Environments

Supplementary Material

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1 Quantitative Evaluation

To quantitatively test all the Aml HMAS functionality, we consider two reference environments with the following setup. *Lab308* is a smart research lab with indoor temperature and light intensity sensors, plus controllable lights and blinds. We build a HomeAssistant deployment using virtual instances of devices in the *Lab308* setup using the HACS integration¹. We showcase both explicit and ambiguous interactions by first asking the Aml HMAS system to bring *Lab308* in a low-light condition (close blinds and turn off lights) and then complain about it "being too dark in here" (an implicit intent of increasing the luminosity in the lab). We further create a second HomeAssistant virtual deployment, whose room and device setup is taken from HomeBench's *Home17* environment, featuring study room lights, climate control devices, and entertainment services.

In these two environments we consider a total of 17 different requests, covering all the following *types*:

- **explicit** requests:
 - single: the request is for a *single*, explicit affordance of a *single*, explicit device
 - multi-command: the request is for multiple individual, explicit affordances of different, explicitly named devices
- **implicit** requests: the request is expressed in an ambiguous or incomplete manner
- **state check**: the request asks for the value of a given device property

The requests are repeated in two phases:

Phase A: Cold start: No signifiers are available. All actionable requests go through the LLM planning path.

Phase B: Warm start: Signifiers extracted from Phase A successes are used for signifier matching. Matched prior experience can be used either directly (for *explicit* requests) or as *affordance hints* that are added to the LLM planning context for *implicit* requests

The list of requests and the way they are handled following each phase is summarized in Table 2.

Table 1 shows the results of request success rates, impossible action detection, number of LLM calls made during reasoning and the average latency of the planning for the request solution for

both Phases: A - cold start (no prior experience), B - warm start (prior experience available from successfully executed plans).

Table 1: Summary results across three evaluation phases. Requests: total requests; Fast/LLM/Imp.: number of requests per planning path; Succ.: success rate; Avg Lat.: average planning latency for actionable (non-impossible) requests; LLM Calls: total LLM API calls.

Phase	Req.	Fast	LLM	Imp.	Succ.	Avg Lat.	LLM Calls
A (Cold)	17	0	14	3	100%	2449 ms	17
B (Warm)	17	9	5	3	100%	337 ms	8

In Phase A, all 14 actionable requests are routed through the LLM, producing valid plans with correct affordance URLs in 17 calls to the LLM reasoning. The 3 impossible requests (no heater in either environment, no brightness control in HomeBench) are correctly identified. Average planning latency is 2449 ms, establishing the baseline for LLM-based planning.

In Phase B, the 14 actionable Phase A requests produce 21 signifiers (14 *Lab308*, 7 *HomeBench*), since multi-intent and implicit requests yield one signifier per action node. With these signifiers, 9 of 14 actionable requests hit the experience reuse (fast) path: zero LLM calls for planning, sub-second latency. This includes the explicit single (e.g. "turn on the light", "set brightness to 80"), and multi-intent requests. The implicit requests (e.g. "it's too dark") still imply an LLM call, because the signifier matches lead to affordance *hints* that get injected in the LLM context window. The limitation of the context window to a subset of affordances instead of all the available ones also leads to a faster planning time.

Request L07, although single and explicit, implies a modification request ("dim the light by 20") and this requires LLM-reasoning to determine the value to which to modify the brightness parameter, given the current environment state. Overall, for cases where experience can be reused, the average planning latency drops from 2449 ms to 337 ms, and LLM calls drop from 17 to 8.

¹Home Assistant Community Store: <https://www.hacs.xyz/>

Table 2: Complete list of evaluation requests grouped by phase and environment. Phase A (cold start) and Phase B (warm start) use all 17 requests; `exp: single` are explicit single requests, `exp: multi` are explicit multi-command requests, `check` are state verification requests. The `imp.` marker in the Path (A) or Path (B) column indicate that the request is actually impossible to solve given the available affordances and the LLM reasoning must detect this as such. The *LLM (hints)* label in Path (B) indicates that Signifier matches are used to inject *affordance hints* in the LLM context window, rather than to reuse an existing plan in the *fast* case. Note that LLM hints are used in the case of implicit requests or when an explicit requests involves a *modification* of state, in which case the LLM needs to reason on the value that the state needs to change to.

ID	Env.	Category	Request text	Path (A)	Path (B)
<i>Phase A & B — Lab308</i>					
L01	Lab308	exp:single	“turn on the light”	LLM	fast
L02	Lab308	implicit	“it’s too dark in here”	LLM	LLM (hints)
L03	Lab308	exp:single	“set brightness to 80”	LLM	fast
L04	Lab308	exp:single	“open the blinds”	LLM	fast
L05	Lab308	implicit	“make the room bright”	LLM	LLM (hints)
L06	Lab308	exp:multi	“turn off the light and close the blinds”	LLM	fast
L07	Lab308	exp:single	“dim the light by 20”	LLM	LLM (hints)
L08	Lab308	check	“is the light on?”	LLM	fast
L09	Lab308	exp:single	“turn on the heater”	imp.	imp.
L10	Lab308	exp:single	“close blinds to 50%”	LLM	fast
<i>Phase A & B — HomeBench Home 17</i>					
H01	Home 17	exp:single	“turn on the study room light”	LLM	fast
H02	Home 17	implicit	“I can’t read anything at my desk”	LLM	LLM (hints)
H03	Home 17	exp:single	“set the light color to warm white”	LLM	fast
H04	Home 17	exp:single	“turn off the light”	LLM	fast
H05	Home 17	implicit	“make the room cozy”	LLM	LLM (hints)
H06	Home 17	implicit	“increase the brightness”	imp.	imp.
H07	Home 17	exp:single	“turn on the heater”	imp.	imp.